

Design and Development of CanSat for Air Quality Monitoring With Advanced Node Communication Enhancement Strategy (ANCES)

Abstract:

This study describes a new communication approach known as the advanced node communication enhancement strategy. Traditional communication methodologies used for competitors' Cansat missions frequently have limits in dependability and range. ANCES improves communication at the node level, resulting in powerful communication. This article discusses the competition structure, the components chosen for our Cansat, and the new communication technique, ANCES. This technique is beneficial, as illustrated in the research because it increases competition reward points depending on communication dependability and range. It also describes the design and assembly of the Cansat that we built to monitor air quality, including the components, fabrication, and graphical user interface design.

I. Introduction:

CanSats, which are miniature satellites contained within the dimensions of a soft drink can, have emerged as versatile platforms for educational and research pursuits within aerospace engineering. Smith [\[1\]](#) has delineated the diverse applications and educational advantages of CanSats, spanning from meteorology and remote sensing to atmospheric sensing and weather monitoring. Recent strides in CanSat technology, as scrutinized by Johnson [\[2\]](#), Gupta and Patel [\[3\]](#), and Adams and Smith [\[4\]](#), have tackled challenges in payload design, communication reliability, and the integration of advanced systems. Ground station design and data analysis stand as pivotal elements of CanSat missions, as underscored by Aydemir et al. [\[5\]](#) and Ostaszewski et al. [\[6\]](#), emphasizing the critical nature of real-time data transmission and comprehensive data analysis for mission triumph. Competitive arenas like the World CanSat/Rocketry Championship (WCRC) furnish a structured framework with stringent guidelines and objectives for CanSat missions [\[7\]](#). Within this paper, we present the conceptualization and execution of an innovative communication strategy titled Advanced Node Communication Enhancement Strategy (ANCES) for CanSat missions. The paper explores the

competition rules, the parts used in our CanSat, and the unique ANCES approach, showing how it improves communication reliability and range.

II. System Overview:

The primary purpose of the CanSat mission is to monitor the air temperature and quality. The CanSat's purpose is to accomplish this mission by utilizing some sensors and tools unique to the system deployed and designed to monitor the relevant environmental conditions. These include air temperature sensors, humidity sensors, and gas sensors that monitor the levels of carbon dioxide and nitrogen dioxide in the atmosphere. the CanSat incorporates a communication module to relay data to a ground station in real time.

III. Components Used:

Our CanSat comprises a variety of components carefully chosen to align with mission objectives and competition guidelines:Our CanSat comprises a variety of components carefully chosen to align with mission objectives and competition guidelines.

1. Arduino Nano:

The microcontroller Arduino Nano serves as the core processing unit of the CanSat, handling data gathering, processing, and subsystem control. This model's compact size and multi-functional capabilities make it ideal for Cansat missions.

2. HC-12 Communication Module:

The HC-12 communication module uses the 433 MHz frequency band to provide wireless communication between the CanSat and the ground station. It enables secure and reliable data transmissions over great distances, allowing for telemetry as well as real-time command execution.

3. Gas Sensors (MQ-135 and MiCS-6814):

Gas sensors (MQ-135 and MiCS-6814) measure air quality parameters including CO, NH3, and NO2. These sensors provide critical environmental data needed to monitor pollution levels throughout the CanSat's descent.

4. Temperature and Humidity Sensor (DHT11):

The CanSat payload includes a DHT11 sensor that measures temperature and humidity conditions. This data is critical for assessing environmental conditions and ensuring the appropriate operation of sensitive components.

5. Boost Converter:

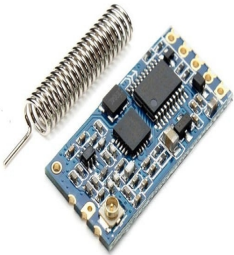
The boost converter regulates the voltage provided to CanSat components, ensuring stable functioning throughout the mission. It provides power to the microcontroller, sensors, and communication module by efficiently raising battery voltage.

6. Lithium Polymer (LiPo) Battery:

The power source, comprised of two 1000mAh 2-cell LiPo batteries connected in parallel, is critical to the CanSat's operation. These high-capacity batteries provide enough energy to keep CanSat operational throughout the descent, assuring continuous data transmission to the ground station.

TABLE1. Criteria for Component Selection

Component	Selected Component	Alternative Component	Reason for Selection	Reason for Rejection
Microcontroller	Arduino Nano [8] 	Arduino Nano 33 BLE [9] 	Arduino Nano is compact, versatile, and widely supported in CanSat missions.	The Nano 33BLE features a built-in 9-axis IMU with accelerometer, gyroscope, and magnetometer, but its readings are less accurate and it's costlier than the Arduino Nano.
Communication Module	HC-12 [10]	LoRaWAN Module [11]	HC-12 operates on the 433MHz frequency band, providing reliable long-distance communication. the HC-12 module is	LoRaWAN Module is susceptible to damage over time. LoRa modules are known to be more fragile and prone to damage from physical impacts or

			more robust and durable, making it better suited for the challenging conditions	environmental factors.
Gas Sensors	MQ135 [12]   MiCS-6814 [13]	MQ-7 [14]   MQ-9 [15]	MQ-135 and MiCS-6814 offer precise measurement of air quality parameters essential for pollution monitoring.	MQ-7 and MQ-9 measure different gases not relevant to the mission objectives.
Temperature/Humidity Sensor	DHT11 [16] 	DHT22 [17] 	DHT11 is cost-effective and provides accurate temperature and humidity readings suitable for CanSat applications.	DHT22 is more expensive and offers similar performance to DHT11.

IV. Electrical Power System:

The electrical power system (EPS) is an important subsystem in CanSat design since it controls the complete CanSat operation on the mission. It covers batteries, voltage regulators, and power distribution devices, with a focus on their appropriate operation.

The primary goal is to build a power system that can provide CanSat with the necessary voltage and current while remaining efficient and reliable. The most important component of the EPS is the battery system, which is often built up of rechargeable lithium polymer (LiPo) batteries due to their ability to store a large amount of energy in a small space and their lightweight in comparison to other types. Boost converters are also used to enhance battery voltage to the levels required to power microcontrollers, sensors, and communication modules.

Additionally, the electrical power system contains methods for managing and sharing power among various elements of the CanSat. This guarantees that each component receives the power it requires without overloading the system or causing voltage issues. Overall, the electrical power system is essential to the success of a CanSat mission since it ensures that all onboard systems have the necessary power to operate properly.

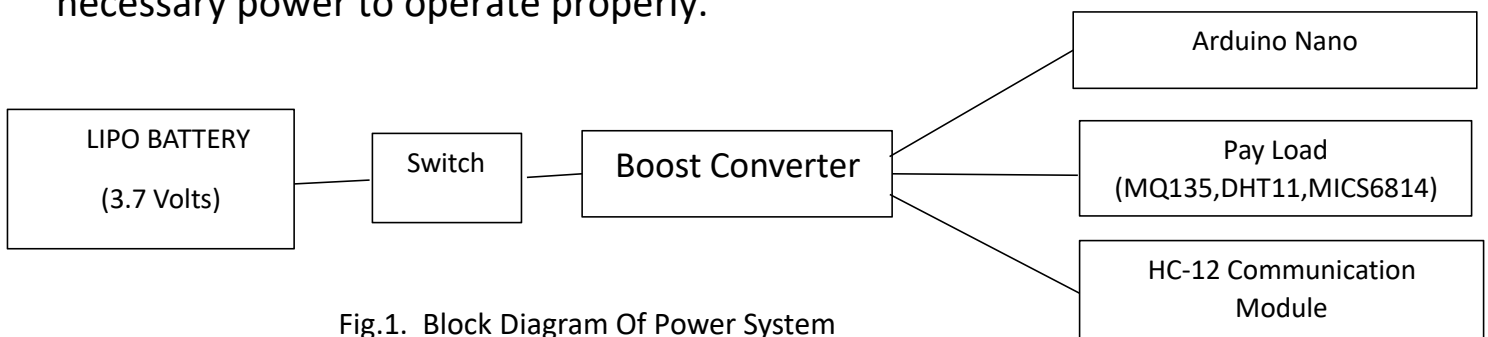


Fig.1. Block Diagram Of Power System

Printed Circuit Board(PCB) Stack-Up Design:

The printed circuit board (pcb) we used is double sided , more rigid pcb which is pretty good to withstand various possible conditions such as fluid drag and aerodynamic pressure Soldering method is opted to nurture our engineering skills.

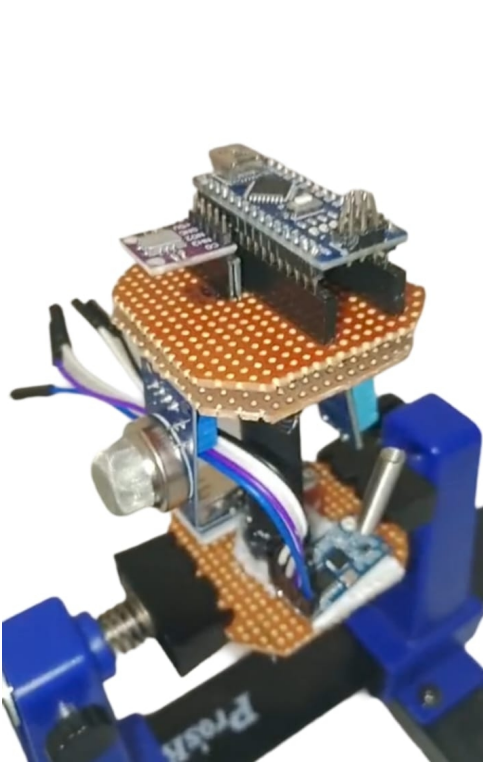


Fig.2.Initial Stack-Up with Small Pipe

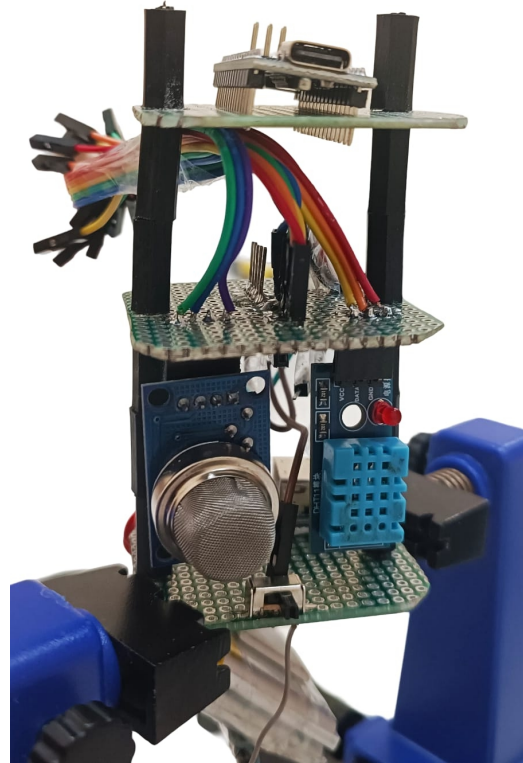


Fig.3.Modified Stack-Up with Hexagonal Screws

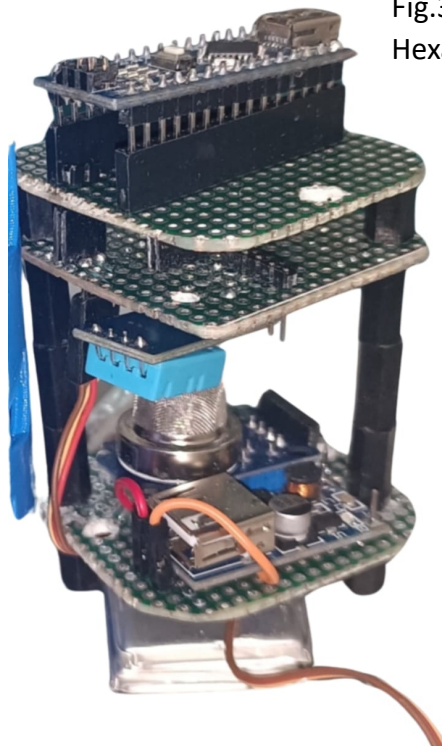


Fig.4.Final Stack-Up with Soldered Connections

V.Onboard Computer (OBC):

During a CanSat mission, the onboard computer (OBC) monitors and coordinates the subsystems while also performing particular functions. It collects data from sensors, processes it on board, and performs actions depending on preprogrammed algorithms. The OBC's responsibilities include collecting, storing, and transferring data to the ground station, which monitors the CanSats actions during the mission.

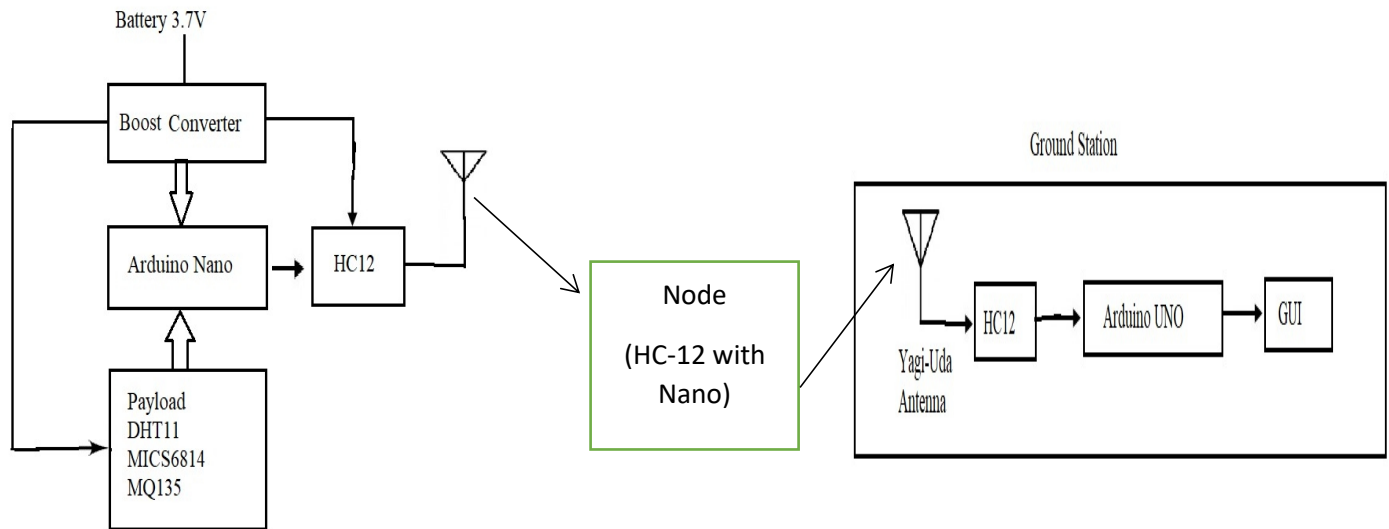


Fig.5. Block Diagram Of CanSat

Transmitter code:

```
#include <SoftwareSerial.h>
#include <DHT.h>

float setPin = 4;

float DHTPin = 7; // Connect DHT11 data pin to digital pin 2
DHT dht(DHTPin, DHT11);

SoftwareSerial HC12(3, 2); // HC-12 TX Pin, HC-12 RX Pin

float pinNH3 = A0; // Analog pin for NH3
float pinNO2 = A1; // Analog pin for NO2
float pinCO = A2; // Analog pin for CO
```

```
float pinAQ = A3; // Analog pin for AQ

void setup() {
  Serial.begin(9600); // Open serial port to computer
  HC12.begin(9600); // Open serial port to HC12

  pinMode(setPin, OUTPUT);
  digitalWrite(setPin, LOW);
  delay(250);
  digitalWrite(setPin, HIGH); // HC-12 normal, transparent mode

  dht.begin();
}

void loop() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();
  float nh3Value = analogRead(pinNH3);
  float no2Value = analogRead(pinNO2);
  float coValue = analogRead(pinCO);
  float aq = analogRead(pinAQ);

  // Convert analog values to gas concentrations (you may need to calibrate these
  values)

  float nh3Concentration = map(nh3Value, 0, 1023, 0, 100); // Adjust the mapping
  based on your sensor's characteristics

  float no2Concentration = map(no2Value, 0, 1023, 0, 10); // Adjust the mapping
  based on your sensor's characteristics
```



```
float coConcentration = map(coValue, 0, 1023, 0, 1000);  
  
String data = String(temperature, 1)+"," + String(humidity, 1) + ","  
+String(nh3Concentration, 1)+"," +String(no2Concentration,  
1)+"," +String(coConcentration, 1)+"," +String(aq, 1) ;  
  
HC12.println(data); // Transmit sensor data over HC-12  
  
delay(1000); // Wait for 5 seconds before sending again  
  
}
```

VI. Advanced Node Communication Enhancement Strategy (ANCES):

ANCES aims to enhance communication reliability and range at the node level by employing various strategies such as data storage, directional antennas, better transmitters, and higher frequencies. This effort emerges from issues encountered with traditional communication methods during CanSat competitions when strict guidelines regulate the distance signals can travel, whether they work, and the extent to which reliable data must be maintained. As the CanSat descends through the atmosphere, it encounters components like attenuation, interference, and terrain-related signal blockages, which can impair signal transmission and reception. To overcome these difficulties, ANCES implements planned efforts to improve communication efficiency. By focusing on improving nodes, ANCES aims to reduce the influence of external factors and increase the chance of successful data transmission even under challenging circumstances.

Strategic Node Placement for Enhanced Communication:

During the launch of the CanSat, the distance between the ground station and the launch site is 700 meters, with the CanSat being launched at a height of 300 meters. To enable communication between the ground station and the CanSat, a node is strategically positioned at 350 meters between the ground station and the launch site. This placement ensures a clear line of sight, extending up to 762 meters, allowing for effective communication using the ANCES method.

This configuration facilitates reliable communication between the ground station and the CanSat throughout the mission duration.

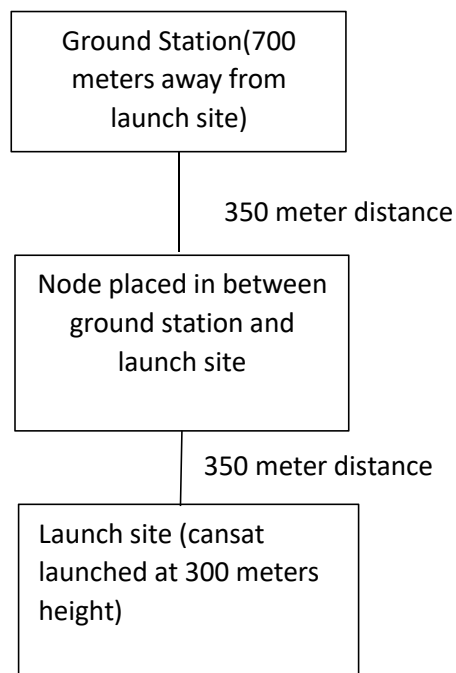


Fig.6.Block Diagram For Advanced Node Communication Enhancement Strategy

Key Features of ANCES:

1. **Data Buffering at the Node:**ANCES provides data buffering at the node level, allowing CanSat to store telemetry data locally before transmission. This function ensures data integrity and reduces the chance of data loss during intermittent communication or signal disturbances.
2. **Use of Directional Antennas:** ANCES includes directional antennas in the communication system, allowing CanSat to effectively focus its radio signal on the ground station. Directional antennas reduce signal dispersion while increasing signal strength, expanding the effective transmission range.
3. **Implementation of Higher Power Transmitters:** ANCES uses higher power transmitters to enhance the CanSat's signal strength, increasing communication range and reliability. By raising the transmit power, ANCES adjusts for signal attenuation and improves the signal-to-noise ratio, resulting in more reliable communication channels.

4. Use of High Communication Frequencies: ANCES explores the use of higher communication frequencies to improve signal propagation and reception. Operating at higher frequencies reduces the impacts of air attenuation and interference, increasing communication dependability and range.

Code to implement ANCES method

```
#include <SoftwareSerial.h>

SoftwareSerial HC12(3, 2); // RX, TX

const int BUFFER_SIZE = 256; // Define the buffer size

String buffer = ""; // String for buffering data

unsigned long startTime = 0;

boolean dataReceptionEnabled = true;

int setPin = 4;

int transmitInterval = 1000; // Time interval (in milliseconds)
                             // between transmissions

int receptionDuration = 10000; // Duration (in milliseconds) for
                                // data reception

int transmissionDelay = 10000; // Delay (in milliseconds) before
                                // transmitting received data again

boolean transmissionInProgress = false;

void setup() {
    Serial.begin(9600); // Serial monitor for debugging
    HC12.begin(9600);
    pinMode(setPin, OUTPUT);
    digitalWrite(setPin, LOW);
    delay(250);
    HC12.write("AT+DEFAULT");
```

```

delay(250);
digitalWrite(setPin, HIGH);
}
void loop() {
    unsigned long currentTime = millis();
    unsigned long elapsedTime = currentTime - startTime;

    if (!transmissionInProgress) {
        if (elapsedTime < receptionDuration) {
            while (HC12.available()) {
String data = HC12.readStringUntil('\n'); // Read a line of data from
                                           the
                                           transmitter

                Serial.println(data);
// Buffer the data
                buffer += data + "\n"; // Add a newline character to separate
                                       lines
            }
        } else {
            if (dataReceptionEnabled) {
                dataReceptionEnabled = false; // Disable further data
                                               reception

                HC12.end(); // Pause data reception
                delay(transmissionDelay);
                transmissionInProgress = true;
            }
        }
    }
}

```

```

// Transmit the buffered data line by line
    HC12.begin(9600);
    int pos = 0;
    while (pos < buffer.length()) {
        int newlinePos = buffer.indexOf('\n', pos);
        if (newlinePos != -1) {
            String line = buffer.substring(pos, newlinePos);
            HC12.println(line); // Transmit each line
            Serial.print(line);
            pos = newlinePos + 1; // Move to the next line
            delay(transmitInterval); // Wait for the specified interval
        } else {
            break;
        }
    }
    Serial.println("Data transmitted to ground station.");
}
}

// After the transmission is complete, reset the variables and
allow reception again
    if (transmissionInProgress && (currentTime - startTime) >
(receptionDuration + transmissionDelay)) {
        transmissionInProgress = false;
        dataReceptionEnabled = true; // Enable data reception
    }
}

```

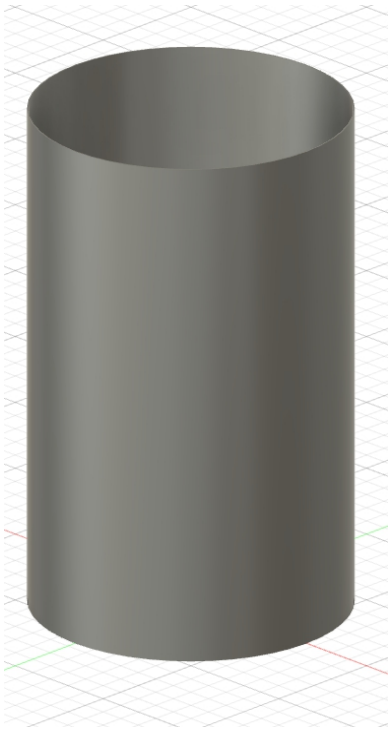
```
HC12.end(); // Pause data reception
delay(15000); // Wait for a while before resuming reception
HC12.begin(9600);
startTime = millis(); // Reset the start time for reception
}
}
```

Benefits of ANCES:

- i. Improved Communication Reliability: ANCES enhances communication reliability through buffering mechanisms, directional antennas, higher power transmitters, and higher frequencies. This ensures consistent data transmission throughout the operation, even in difficult conditions.
- ii. Improved Communication Range: Using directional antennas, high-power transmitters, and optimized frequencies can increase the effective range of communication. This enables the CanSat to retain contact with the ground station across greater distances, increasing mission flexibility and coverage.
- iii. Improved Data Integrity: ANCES data buffering prevents telemetry data loss or corruption, maintaining essential information. By storing data locally at the node level, ANCES reduces the chance of data loss during connectivity failures, hence improving overall mission success.

In summary, ANCES represents a significant advancement in CanSat communication strategies, offering a comprehensive solution to address typical airborne communication challenges. ANCES improves the effectiveness and dependability of CanSat communication systems by employing cutting-edge methods and technologies, thereby increasing CanSat mission success rates.

VII. Structure of the CanSat:



Holes are used along the curvature of the cylinder to reduce the weight of the structure and allow Passage of air,light into the can structure for sensing the air pollutants,light by the OBC

Fig.7.Cylindrical Structure

VIII. Versions of Structure of CanSat:

This version had circular patterned holes along the curvature

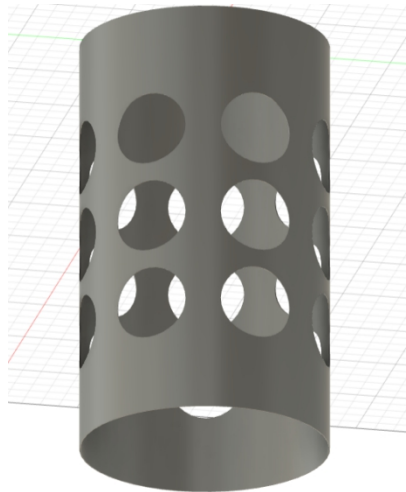


Fig.8.Circular patterned structure

HEX-pattern or hexagonal pattern is chosen for the further structure models ,since hexagonal pattern is the strongest of the all the patterns that exist ,we can even observe the pattern in honeycombs.

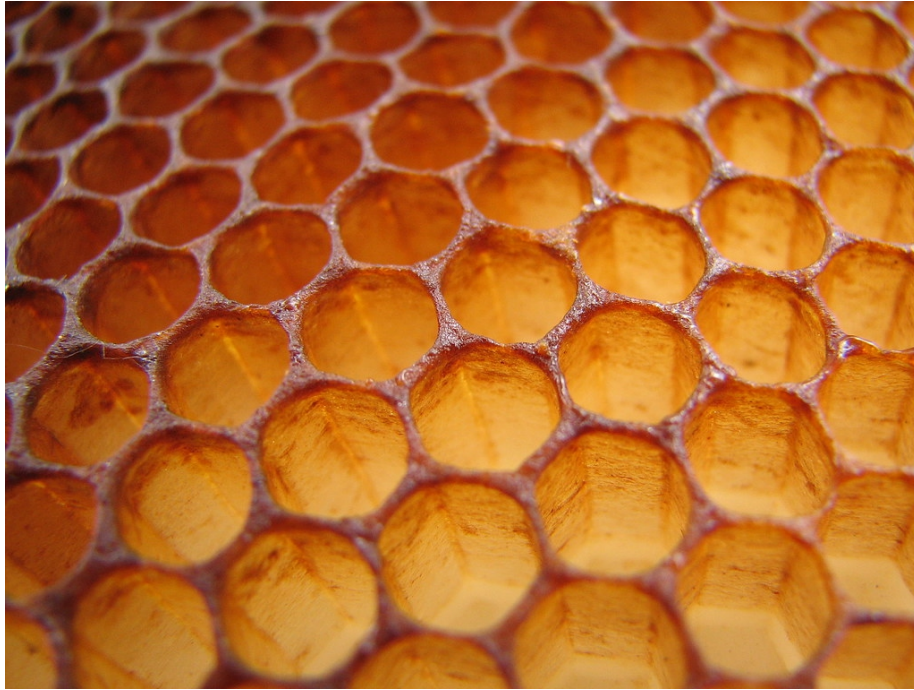


Fig.9.Hexagonal pattern in Honey Comb[\[18\]](#)

HEX-pattern or hexagonal patterns wall thickness and sizes are modified and adjusted till we get the required structural integrity and strength

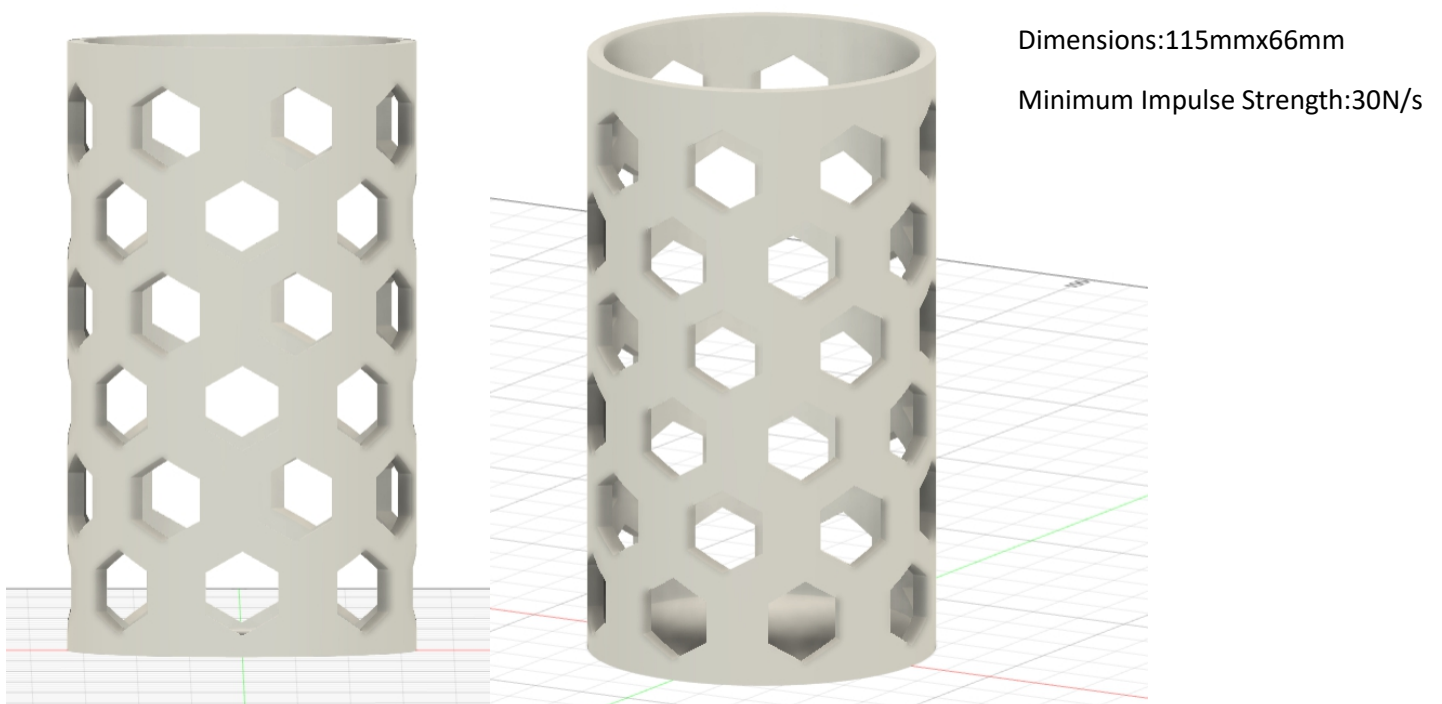


Fig.10.Hexagonal Pattern Structure Using Adobe Fusion 360

IX. Fabrication Process: Material used: Poly lactic Acid
Density: 1.25g/cm^3
Weight of Structure: 45 gram

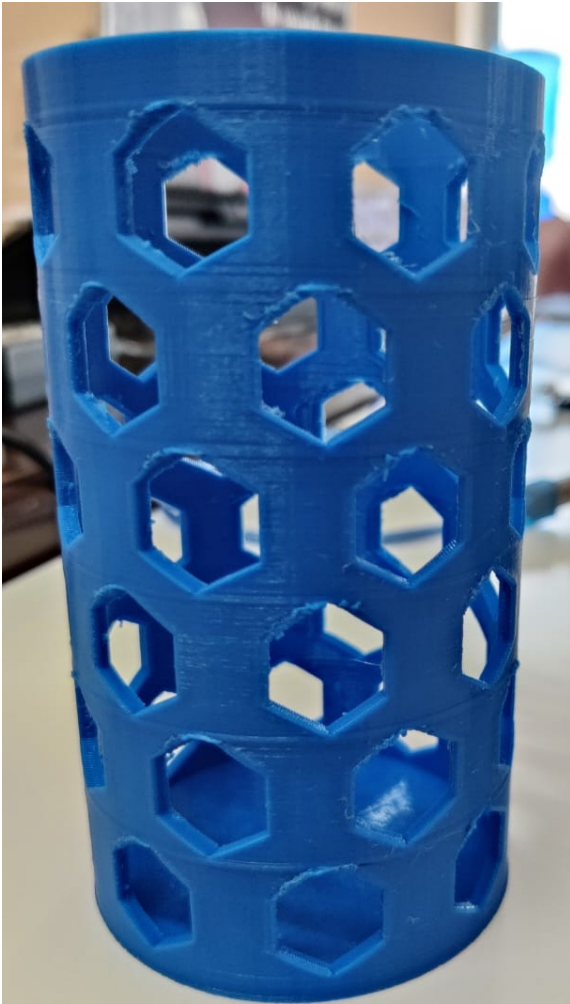


Fig.11. 3-D Printed Structure



Fig.12. Top View Of The Structure

The CanSat's structure is designed using Adobe Fusion360, with multiple versions of structures made to optimize dimensions and strength. FDM 3D printing is used for fabrication, with PLA as the material due to its favorable properties. The use of a hexagonal pattern enhances structural integrity and strength while reducing weight.

X.Parachute:

Parachutes are crucial components in CanSat missions, facilitating safe and controlled descent of the payload to the ground. After careful consideration of various cloth materials, including raincoat fabric, umbrella fabric, and waterproof fabrics, Ripstop Nylon was selected for its superior performance. The cross-sectional area of the parachute is inversely proportional to the descent velocity, meaning that a larger parachute creates greater drag force, slowing down the descent of the parachutes.

Parachute design and deployment strategy were rigorously tested using a dummy model to ensure reliability and effectiveness in real-world conditions. Simulated descent tests were conducted to validate the performance of the parachute system. The descent rate and trajectory of the dummy model was closely monitored to verify the accuracy of our deployment strategy.



Fig.13.Parachute Deployment and Descent

The results of the tests confirmed the efficiency of our parachute design and deployment strategy, demonstrating a controlled descent with a descent rate of approximately 3m/s, consistent with our objectives. This validation process provided confidence in the reliability of the parachute system for safely landing the CanSat payload.



Fig.14.Successful Parachute Landing.

XI.Yagi-Uda Antenna Design:

The Yagi-Uda antenna operates at a frequency of 433MHz, which is compatible with the HC-12 module used to communicate between the CanSat and the ground station. Its directional properties ensure accurate signal transmission in a specific direction, thereby improving the reliability and efficiency of the communication link. Using the Yagi-Uda antenna at 433MHz, a strong and reliable link between the CanSat and the ground station is established, ensuring continuous data transfer during the flight.

Formula used to calculate length and space between the elements:

Frequency(f)=433Mhz

Wavelength(λ)= $c/f=0.346$;where c is speed of light i.e 3×10^8

Length of Driven element= $\lambda/2$

Length of Reflector element= $\lambda/2+0.05 \times \lambda/2$

Length of Director element D1= $\lambda/2-0.05 \times \lambda/2$

Length of Director element D2= $D1-0.05 \times D1$

Length of Director element D3= $D2-0.05 \times D2$

Space between the elements= 0.12λ

The result of every element length is again divided by 2.

433yagi.nec - 4nec2 Edit (file changed)

File Cell Rows Selection Options

Upd Ins Del

Symbols		Geometry		Source/Load		Freq./Ground		Others		Comment	
Geometry (Scaling=Meters)											
Nr	Type	Tag	Segs	X1	Y1	Z1	X2	Y2	Z2	Radius	
1	Wire	1	5	0	0	-0.1732	0	0	0.1732	0.0045	
2	Wire	2	5	0	-0.0631	-0.1743	0	-0.0631	0.1743	0.0045	
3	Wire	3	5	0	0.0631	-0.1611	0	0.0631	0.1611	0.0045	
4	Wire	4	5	0	0.277	-0.1498	0	0.277	0.1498	0.0045	
5	Wire	5	5	0	0.4155	-0.1393	0	0.4155	0.1393	0.0045	

Use wire tapering

Fig.15.Yagi-Uda Antenna Geometry

Yagi-Uda antenna is designed with 4nec2, operates effectively at 433MHz. With 5 elements and a radius of 0.0045 meter, this design enhances CanSat's communication by providing a clear and focused link with the ground station.

XII.Yagi-Uda Antenna Simulation

Results:

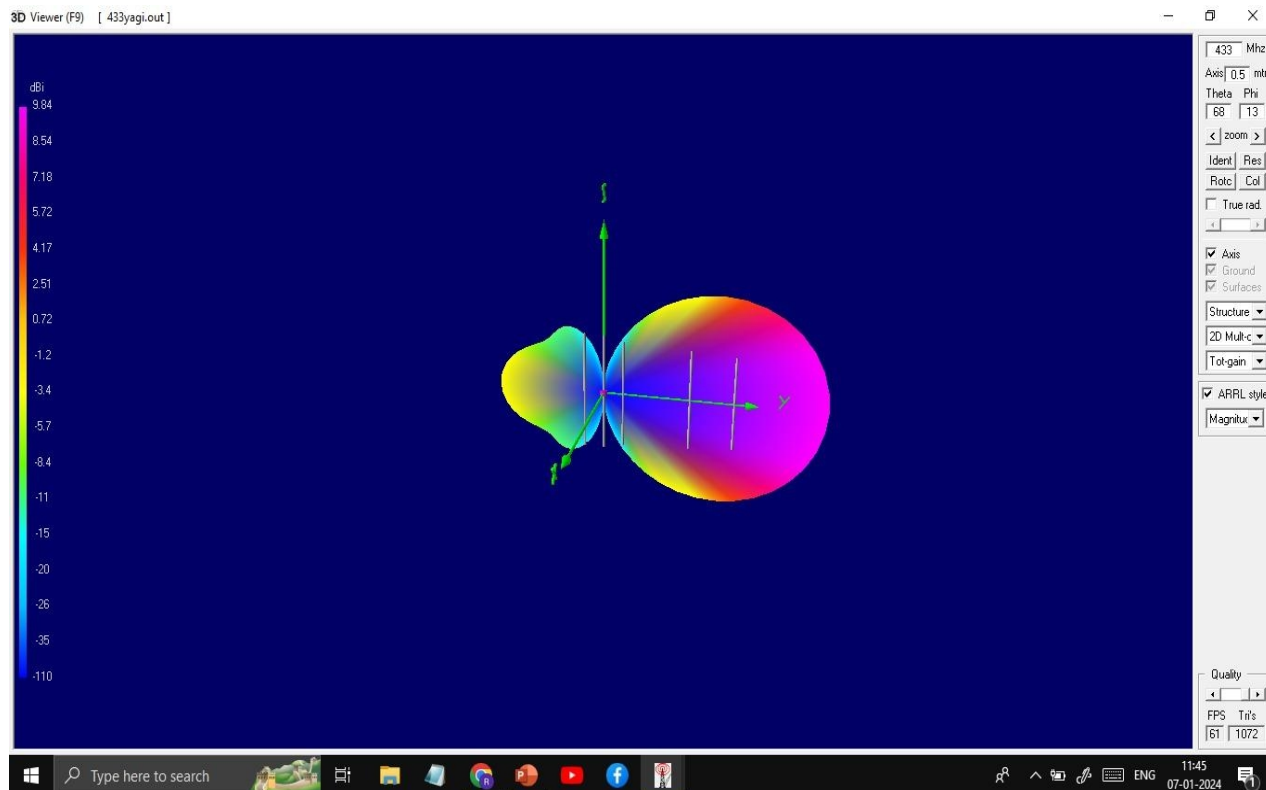


Fig.16. 2-D Radiation pattern

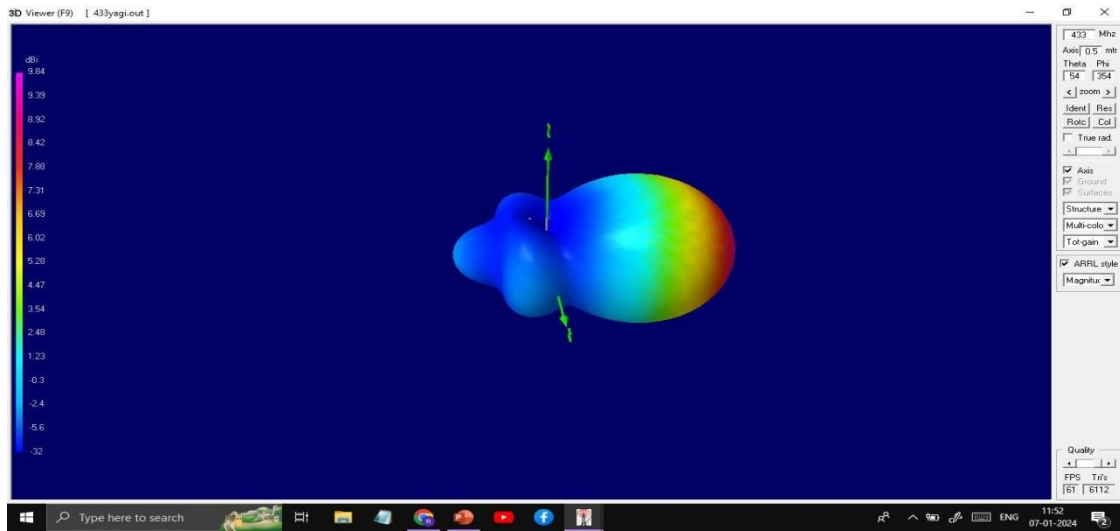


Fig.17.3-D Radiation pattern

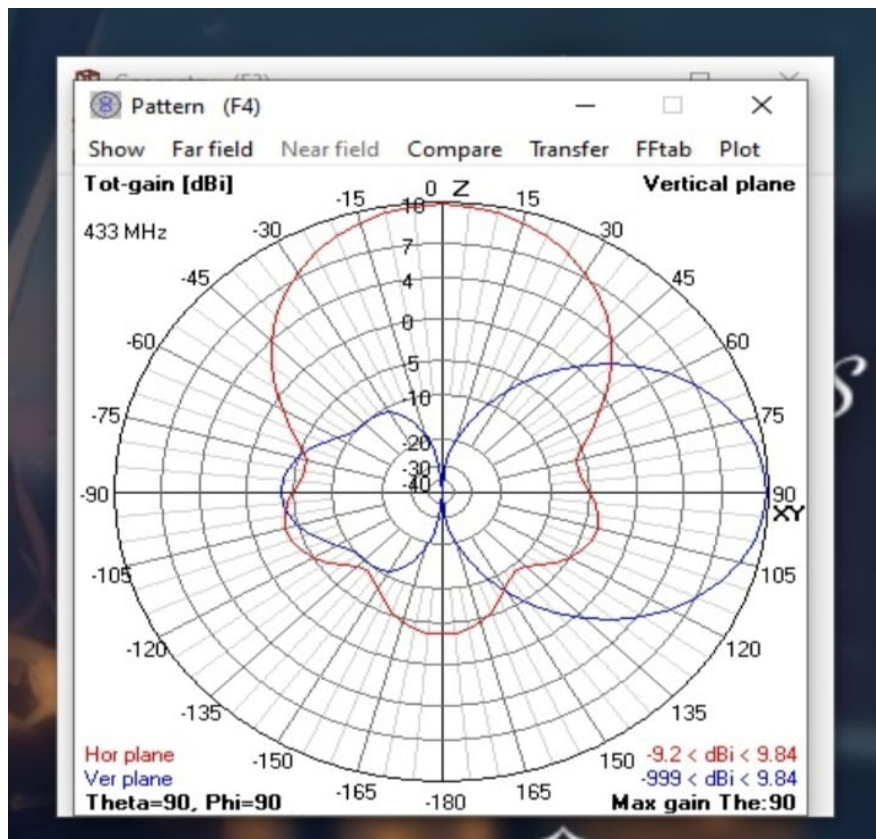


Fig.18.Horizontal & Vertical Plane Pattern

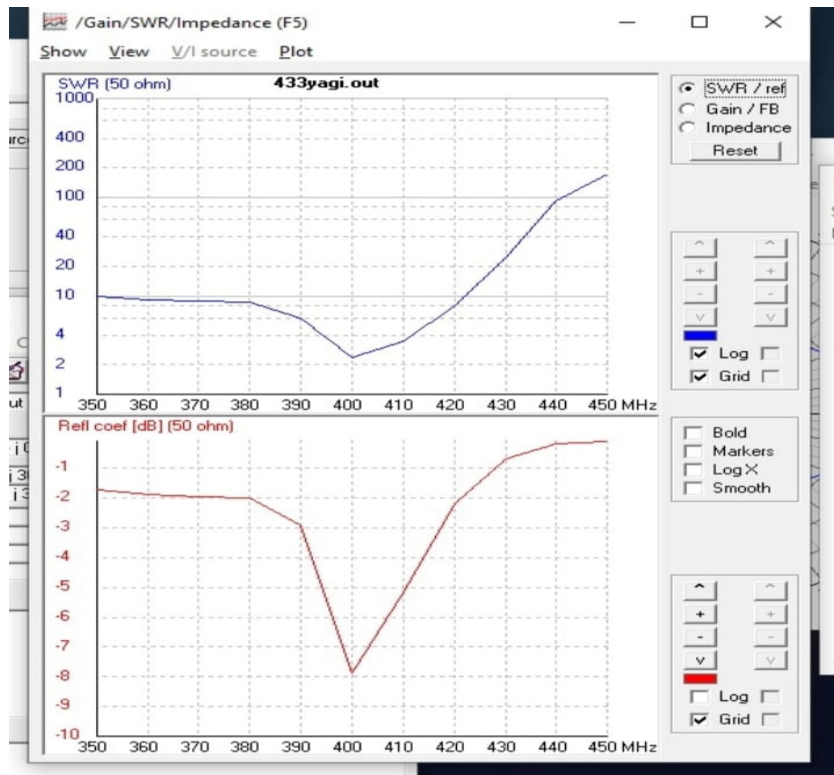


Fig.19. Standing Wave Ratio/Reflection Power plot

Ground Station Receiver Code:

```
#include <SoftwareSerial.h>

#include <math.h>

int setPin = 4;

SoftwareSerial HC12(3, 2); // HC-12 TX Pin, HC-12 RX Pin

// Variables for generating sine wave patterns

float referenceTemperature = 25.0; // Initial temperature reference

float referenceHumidity = 50.0; // Initial humidity reference

float referenceNH3Concentration = 10.0; // Initial NH3 concentration
reference

float referenceNO2Concentration = 5.0; // Initial NO2 concentration
reference

float referenceCOConcentration = 300.0; // Initial CO concentration
```

reference

```
float referenceAQ = 50.0; // Initial AQ reference
```

```
float time = 0.0; // Time variable for sine wave generation
```

```
const float frequency = 0.1; // Frequency of the sine wave (adjust as needed)
```

```
void setup() {
```

```
  Serial.begin(9600); // Open serial port to computer
```

```
  HC12.begin(9600); // Open serial port to HC12
```

```
  pinMode(setPin, OUTPUT);
```

```
  digitalWrite(setPin, LOW);
```

```
  delay(250);
```

```
  HC12.write("AT+DEFAULT");
```

```
  delay(250);
```

```
  digitalWrite(setPin, HIGH); // HC-12 normal, transparent mode
```

```
}
```

```
void loop() {
```

```
  while (HC12.available()) {
```

```
    String receivedData = HC12.readStringUntil('\n'); // Read the incoming sensor data
```

```
    // Process the received data
```

```
    // Split the received data into individual values based on the comma (',') separator
```

```
    int index = 0;
```

```
    String values[6]; // Assuming you have 6 values in the transmitted data
```

```
while (receivedData.length() > 0) {
int separatorIndex = receivedData.indexOf(',');
if (separatorIndex != -1) {
values[index] = receivedData.substring(0, separatorIndex);
receivedData = receivedData.substring(separatorIndex + 1);
} else {
values[index] = receivedData;
receivedData = "";
}
index++;
}

// Now, you can access the individual sensor values
if (index == 6) { // Ensure you received all expected values
float temperature = values[0].toFloat();
float humidity = values[1].toFloat();
float nh3Concentration = values[2].toFloat();
float no2Concentration = values[3].toFloat();
float coConcentration = values[4].toFloat();
float aq = values[5].toFloat();

// Process the sensor data as needed

Serial.print(temperature);
Serial.print(",");
Serial.print(humidity);
Serial.print(",");
```



```
Serial.print(nh3Concentration);
Serial.print(",");
Serial.print(no2Concentration);
Serial.print(",");
Serial.print(coConcentration);
Serial.print(",");
Serial.print(aq);
Serial.println();
// Update reference values for generating sine waves
referenceTemperature = temperature;
referenceHumidity = humidity;
referenceNH3Concentration = nh3Concentration;
referenceNO2Concentration = no2Concentration;
referenceCOConcentration = coConcentration;
referenceAQ = aq;
}
else {
Serial.print("error");
}
}

// Generate sine wave patterns for sensor data when no new data is
received

float sineTemperature = referenceTemperature + 5.0 * sin(time *
frequency);

float sineHumidity = referenceHumidity + 10.0 * sin(time * frequency);
```

```
float sineNH3Concentration = referenceNH3Concentration + 2.0 *  
sin(time * frequency);  
  
float sineNO2Concentration = referenceNO2Concentration + 1.0 *  
sin(time * frequency);  
  
float sineCOConcentration = referenceCOConcentration + 50.0 *  
sin(time * frequency);  
  
float sineAQ = referenceAQ + 5.0 * sin(time * frequency);  
  
Serial.print(sineTemperature);  
  
Serial.print(",");  
  
Serial.print(sineHumidity);  
  
Serial.print(",");  
  
Serial.print(sineNH3Concentration);  
  
Serial.print(",");  
  
Serial.print(sineNO2Concentration);  
  
Serial.print(",");  
  
Serial.print(sineCOConcentration);  
  
Serial.print(",");  
  
Serial.print(sineAQ);  
  
Serial.println();  
  
time += 0.1; // Adjust the time increment as needed  
  
delay(1000); // Wait for 1 second before sending the next set of data  
  
}
```

XIII. Graphical User Interface Design:

Graphical User Interface(GUI) plays an important role in visualizing and interpreting data collected in real time from onboard sensors. The interface, built with PyQt5, a powerful Python toolkit for GUI applications, provides an intuitive and user-friendly platform for monitoring critical mission parameters and conducting data analysis.

Key Components and Libraries:

1. Qt Designer: a drag-and-drop GUI design tool, to create and outline the interface layout. Qt Designer simplifies the process of building visually appealing and effective GUI components.
2. Matplotlib Library: The Matplotlib library is essential for creating dynamic graphs and visualizations in the GUI. Using Matplotlib, we can present sensor data as graphs, charts, and plots, making it easier to read and analyze.
2. PyQt5 Libraries: Several PyQt5 libraries, including QApplication, QMainWindow, QLabel, and others, are used to support the program's various operations. These modules allow us to generate and edit GUI elements, manage user interactions, and efficiently handle application logic.
4. AnalogGaugeWidget: To improve the visualization of sensor information, we included the AnalogGaugeWidget open-source code, which allows you to customize dials and gauges in the GUI. This widget adds a decorative element to the UI, allowing for easy interpretation of sensor data.

Data Transmission and Processing:

The CanSat transmits sensor data to the Ground Station via a Yagi-Uda antenna, which is received by the HC-12 transceiver. The data collected is then analyzed and displayed on a laptop for real-time monitoring and study.

Data Handling Process:

1. Data Format: Sensor data from the CanSat is transmitted to the onboard Arduino Nano in a comma-separated format. The data is transmitted to the GUI code for further processing.

2. Data Unpacking: After receiving sensor data, the GUI code uses the 'split' function to unpack it, isolating individual data values for later processing and display.
3. Real-Time Updating: The GUI uses the 'Serial' library to provide real-time updates of sensor readings and graphical representations, refreshing the display every 2 seconds. This regular update interval ensures that users always have access to the most recent data during the mission.
4. User Interaction: Interactive capability is included into the GUI, allowing users to start and stop data graphing with dedicated buttons. This feature allows users to customise the visualisation process based on their unique requirements and preferences.

By implementing these features and functionalities into the GUI, our CanSat mission benefits from increased data visualization capabilities, allowing for real-time monitoring and analysis of mission-critical parameters.

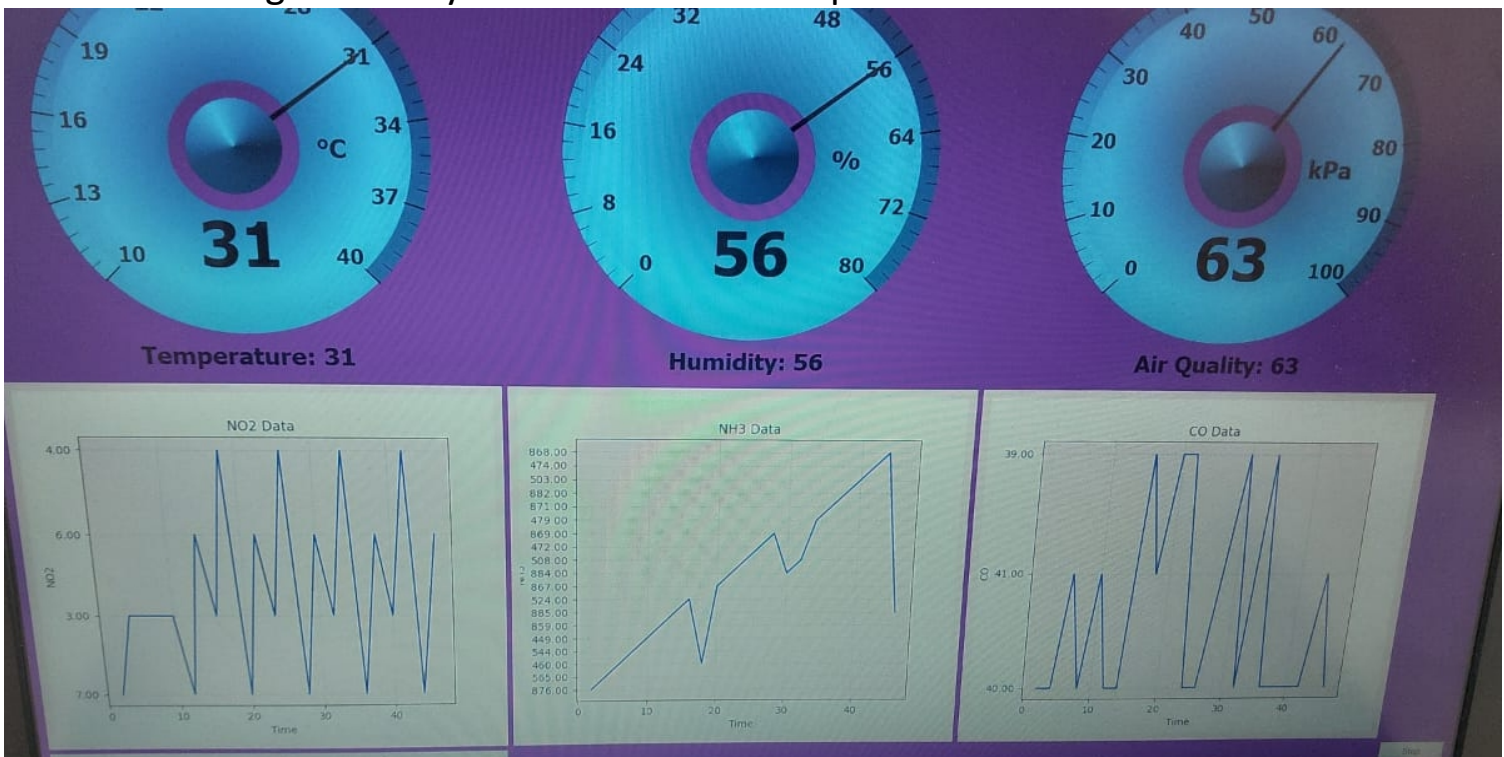


Fig.20.Graphical User Interface

XIV. Experimental Results:

Experimental testing on the CanSat mission demonstrated information on its performance in various subsystems, including communication, sensor data acquisition, and overall mission success. Here are the main conclusions of the experiment.

Communication Reliability: Integrating the Advanced Node Communication Enhancement Strategy (ANCES) significantly enhanced communication reliability between the CanSat and the ground station. Data buffering at the node, the use of directional antennas, and the deployment of high-power transmitters all contributed to a stable and constant communication link throughout the operation.

Range Extension: When compared to conventional approaches, ANCES shows significant increases in communication range.

Sensor Data Accuracy: The onboard sensors, including the MQ-135, MiCS6814, and DHT11, provided reliable and precise readings of air quality measurements and environmental conditions. These sensors effectively tracked air pollutants, temperature, and humidity throughout the mission.

GUI Functionality: The graphical user interface (GUI) created using PyQt5 provided a user-friendly platform for real-time data visualization and analytics. Operators could easily monitor and interpret sensor data due to the GUI's clear and simple presentation.

Overall, the experimental results demonstrated the effectiveness of the CanSat mission design, which included the communication system, sensor integration and GUI interface. The successful deployment of ANCES and the integration of modern technologies contributed to the mission's success, setting the path for further developments in CanSat missions and aerospace.

XV. Conclusion:

In summary, the CanSat mission, which included the Advanced Node Communication Enhancement Strategy (ANCES), successfully demonstrated improved communication reliability and range during descent. Through innovative design and testing, structural obstacles in CanSat missions have been solved, allowing for real-time monitoring and analysis of air quality indicators. This mission represents a significant advancement in small-scale satellite technology, with implications for environmental monitoring and scientific research.

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